

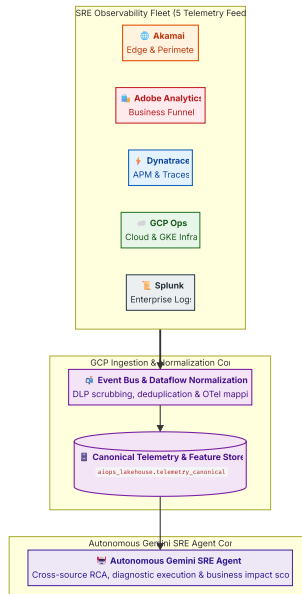
SRE Observability Fleet - Source Telemetry Matrix

Target Platform: Google Cloud Platform (GCP) System of Record: ServiceNow ITSM Version: 2.0 (Production Blueprint) Generated: August 2026

SRE Observability Fleet - Source Telemetry Matrix

1. Executive Overview

The **Autonomous Gemini SRE Agent** correlates telemetry across 5 core observability tools used by enterprise SRE teams. Instead of viewing these tools as isolated monitoring silos, the AIOps platform models them as high-dimensional feature feeds that map to specific operational and business domains:



2. Comprehensive Tool Telemetry Profiles

2.1 Akamai (Edge & Perimeter Observability)

- **Domain:** Edge CDN, DNS, Web Application Firewall (WAF), DDoS mitigation, and Bot Management.
- **Emitted Telemetry:**
 - Edge HTTP latency and Time to First Byte (TTFB).
 - HTTP status distributions (502/503/504 edge vs. origin error rates).
 - WAF security triggers, DDoS mitigation vectors, and Bot Manager classification scores.
- **Ingestion Channel:** Akamai **DataStream 2** HTTPS Push to Cloud Run Ingress → `telemetry.akamai.raw`.
- **Agent Reasoning Value:**
 1. Distinguishes regional ISP/CDN network degradation from core backend microservice crashes.
 2. Isolates bot-driven traffic spikes to prevent unnecessary container autoscaling.

2.2 Dynatrace (APM, Distributed Tracing & Topology)

- **Domain:** Full-stack Application Performance Monitoring (APM), code-level distributed tracing (PurePath), runtime dependency mapping (Smartscape), and Davis AI problem detection.
- **Emitted Telemetry:**
 - Distributed trace spans with OpenTelemetry headers (`trace_id` , `span_id`).
 - JVM/CLR thread pool saturation, Garbage Collection (GC) pauses, and memory heap metrics.
 - Smartscape entity topology DAGs and Davis AI root cause problem webhooks.
- **Ingestion Channel:** Real-time problem webhooks → `telemetry.dynatrace.raw` ; scheduled REST API sync for Smartscape DAGs (`/api/v2/entities`).
- **Agent Reasoning Value:**
 1. Directly identifies the failing class, method, or SQL database lock from PurePath stack traces.
 2. Traverses Smartscape dependency graphs to calculate upstream and downstream blast radius.

2.3 Google Cloud Operations Suite (Cloud & Platform Infrastructure)

- **Domain:** Native GCP infrastructure, Google Kubernetes Engine (GKE), serverless runtimes, and Cloud Audit Logs.
- **Emitted Telemetry:**
 - Node/Pod CPU and Memory saturation, container `CrashLoopBackOff` restart counts.
 - Cloud Pub/Sub unacknowledged message age and Dataflow system watermarks.
 - Cloud Audit Logs (identifying recent CI/CD deployments and configuration changes).
- **Ingestion Channel:** Zero-compute **Cloud Logging Log Router Sinks** → `telemetry.gcp.raw` ; Google Managed Service for Prometheus (GMP).
- **Agent Reasoning Value:**
 1. Correlates sudden application error spikes with recent Kubernetes deployments or configuration drift.
 2. Evaluates infrastructure resource starvation (OOM kills, CPU throttling).

2.4 Splunk (Enterprise Logging & SIEM Forensics)

- **Domain:** Enterprise centralized log repository, physical Point-of-Sale (POS) logs, middleware message queues (Kafka, IBM MQ), and SIEM events.
- **Emitted Telemetry:**
 - Unstructured application logs, stack traces, and database connection pool exceptions.
 - POS terminal hardware and peripheral transaction logs.
 - Splunk Enterprise Security (ES) notable security events.
- **Ingestion Channel:** Splunk **HTTP Event Collector (HEC)** → `telemetry.splunk.raw` ; on-demand REST API for time-windowed SPL queries (`$\pm 10\text{ minutes}$`).
- **Agent Reasoning Value:**
 1. Performs semantic error log clustering across dozens of microservices.
 2. Pulls forensic log context matching the incident's `trace_id` for inclusion in ServiceNow work notes.

2.5 Adobe Analytics (Digital Experience & Business Telemetry)

- **Domain:** Customer conversion funnels, e-commerce checkout telemetry, and digital revenue transactions.
- **Emitted Telemetry:**
 - Real-time Orders Per Minute (OPM), Cart Additions (`scAdd`), Checkout Funnel Drop-offs (`scCheckout`).
 - Payment gateway third-party failure responses.
 - User cohort attributes (geographic region, device type, app version).
- **Ingestion Channel:** Adobe Experience Platform (AEP) Streaming Ingestion → `telemetry.adobe.raw` .
- **Agent Reasoning Value:**
 1. **Silent Outage Detection:** Identifies frontend or payment bugs where technical infrastructure appears green but business transactions are failing.
 2. **Financial Impact Scoring:** Calculates real-time estimated revenue loss (USD/min) to drive accurate incident prioritization (P1 vs. P3) in ServiceNow.

3. Canonical Field Mapping Matrix (`telemetry_canonical`)

All 5 feeds are standardized into the OpenTelemetry-aligned canonical schema in BigQuery:

Canonical Field	Type	Akamai Mapping	Dynatrace Mapping	GCP Ops Mapping	Splunk Mapping	Adobe Mapping
<code>event_id</code>	STRING	<code>requestId</code>	<code>ProblemID</code> / <code>spanId</code>	<code>insertId</code>	<code>_cd</code> / <code>eventId</code>	<code>eventId</code>
<code>timestamp</code>	TIMESTAMP	<code>startEpoch</code>	<code>StartTime</code>	<code>timestamp</code>	<code>_time</code>	<code>time</code>
<code>source_tool</code>	STRING	<code>'akamai'</code>	<code>'dynatrace'</code>	<code>'gcp_ops'</code>	<code>'splunk'</code>	<code>'adobe'</code>
<code>severity</code>	STRING	Derived from HTTP status	<code>ProblemSeverity</code>	<code>severity</code>	Derived from log level	Derived from log level
<code>entity.service_name</code>	STRING	<code>property_name</code>	<code>ImpactedEntity</code>	<code>resource.labels.container_name</code>	<code>source</code> / <code>service</code>	<code>Funnel</code> / <code>check</code>
<code>entity.host</code>	STRING	<code>edge_ip</code>	<code>HostName</code>	<code>resource.labels.node_name</code>	<code>host</code>	N/A (Custom Browser)
<code>entity.cmdb_ci_id</code>	STRING	CMDB CI mapping	CMDB CI mapping	CMDB CI mapping	CMDB CI mapping	CMDB CI mapping
<code>log_payload.trace_id</code>	STRING	<code>X-Trace-ID</code>	<code>PurePathTraceId</code>	<code>trace</code>	<code>trace_id</code>	<code>trace_id</code>
<code>log_payload.message</code>	STRING	Request URI & Error	Error message	<code>textPayload</code> / <code>jsonPayload</code>	<code>_raw</code>	N/A
<code>business_context.orders_per_minute</code>	FLOAT64	N/A	N/A	N/A	N/A	<code>orders_per_minute</code>
<code>business_context.estimated_revenue_loss_usd</code>	FLOAT64	N/A	N/A	N/A	N/A	Calculated

4. Cross-Domain Architectural Linkages

- [Ingestion Master Blueprint & Streaming Pipelines](#)
- [Data Contracts & Canonical Schemas](#)

- [Unified Lakehouse & AI Feature Store](#)
- [Autonomous Gemini SRE Agent Architecture](#)